TAlpha user manual

v. 4.2

Detailed User Guide

Alpha-decay half-life calculations, calculation history, and interactive nuclide diagrams.

Purpose. This guide describes the TAlpha v.4.2 interface. The available nuclide range and α-database version may change after a database update.

General (main) calculator screen

The main screen is used to enter alpha-decay parameters manually and run a calculation. It also accepts values selected on an interactive decay scheme or loaded from the calculation history.

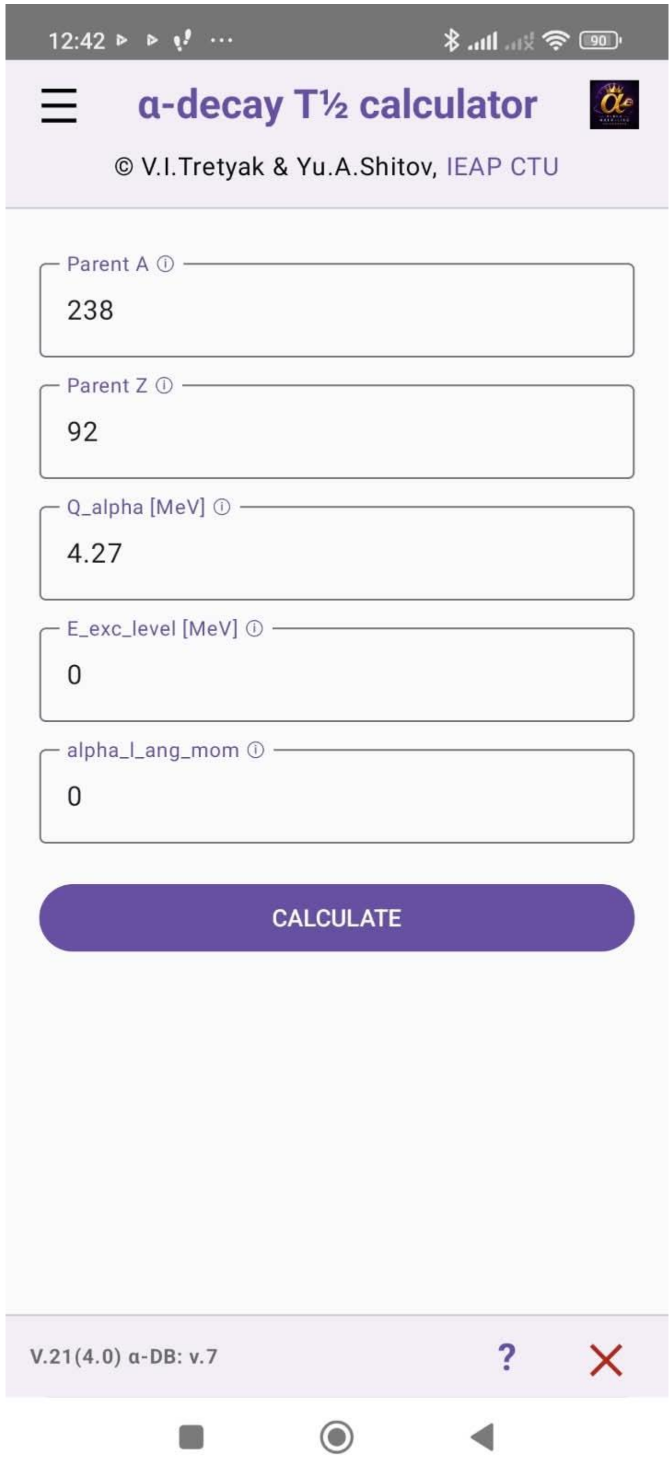

Figure 1. TAlpha main screen: input fields, the CALCULATE button, and the bottom bar.

Top bar

☰ Menu. Opens the side panel containing history commands, nuclide diagrams, logo settings, Premium, and reference information.

Application title. The "α-decay T½ calculator" title identifies the current working screen.

Logo on the right. The standard TAlpha logo. A personal logo does not replace it: when the Premium feature is enabled, the personal logo is shown separately in the center of the bottom bar.

The authors and their institution are shown below the title. The IEAP CTU link opens the associated institute page when a browser is available on the device.

Input fields

Field	What to enter
Parent A	Mass number A of the parent nuclide—a positive integer.
Parent Z	Atomic number Z of the parent nuclide—a positive integer. The A, Z pair should describe a physically existing or investigated nuclide.
Q_alpha [MeV]	Ground-state-to-ground-state alpha-decay energy Qα, in MeV.
E_exc_level [MeV]	Excitation energy of the selected daughter level, in MeV. Enter 0 for a transition to the ground state.
alpha_l_ang_mom	Orbital angular momentum l of the emitted alpha particle—a non-negative integer. Do not enter the level spin J here.

Help icons ⓘ. The small icon next to a field name opens a short explanation of that parameter. Tap outside the hint or use the system Back action to close it.

Physical condition. For the selected branch, the available energy is Qbranch = Qα − Eexc. The calculation requires Qbranch > 0. If the excitation energy is equal to or greater than Qα, the transition is energetically forbidden and the application reports an error.

Entering and validating values

1. Tap the required field. You can select the previous value and replace it using the on-screen keyboard.
2. Enter A, Z, Qα, Eexc, and l. For a transition to the ground state, use Eexc = 0; for a transition with no angular-momentum barrier, use l = 0.
3. Check the units: Qα and Eexc are entered in MeV, not keV.
4. Tap CALCULATE. If an entry is invalid or incomplete, correct the field indicated by the application message.

Important. Values inserted from the database can be edited manually. The associated experimental EXP block remains attached as a reference to the selected nuclide/transition, so compare its parameters with the current Qα and Eexc values before interpreting the result.

Bottom bar

V.<code> (<name>). The version code and version name of the installed application.

α-DB: v.<db>. The version of the loaded nuclide database. This label appears only after the database has been loaded successfully.

?. Opens application help/documentation.

X. Closes the application.

Calculations using manually entered data are available independently of the diagrams. Chart of α-decaying nuclei requires a loaded α-database; an internet connection may be needed for the first download or an update check.

Calculations

After you tap CALCULATE, the results appear below the button in separate cards. Scroll down if all cards do not fit on the screen at the same time.

Figure 2. Result cards for the two parameterizations and the experimental EXP block.

Calculated-result cards

PRC 79 (2009) 054614. Calculation using the Denisov and Khudenko parameterization, Phys. Rev. C 79, 054614 (2009), including the published erratum.

PRC 92 (2015) 014602. Calculation using the updated Denisov parameterization, Phys. Rev. C 92, 014602 (2015).

Each card presents the result in three equivalent forms:

- log10(T½, sec) — the base-10 logarithm of the half-life in seconds;
- T½ in seconds;
- T½ in years, with one year defined as 365.25 days.

Scientific notation such as 2.64e+07 means 2.64×10^7. A difference between the two cards reflects the different phenomenological parameterizations and is not a rounding error.

Experimental EXP card

The EXP card is displayed only when the nuclide database contains experimental information for the selected nuclide or level. The heading shows the branch Eexc and Qα values; the experimental half-life T½exp in years is shown below.

- EXP — a reference experimental value from normalized ENSDF data;
- Eexc — the energy of the daughter-nucleus level associated with the experimental branch;

- Qα — the energy of the specific branch, i.e., the ground-state Qα minus Eexc;
- T½exp — the experimental half-life of the parent nuclide/branch, if defined in the database.

Comparison. First make sure that Qα, Eexc, and l in the input fields match the EXP-card parameters. After a manual edit, the card is intentionally retained as a reference.

History after a calculation

A successful calculation is added to the current session history. The record contains the original A, Z, Qα, Eexc, and l values, the results of both models, and, when available, the experimental Qexp, Elevel,exp, and Texp values. You can then view, save, load, or share the history from the menu.

Menu

Tap ☰ in the upper-left corner to open the side panel. To close it, tap the dimmed area, tap ☰ again, or use the system Back action.

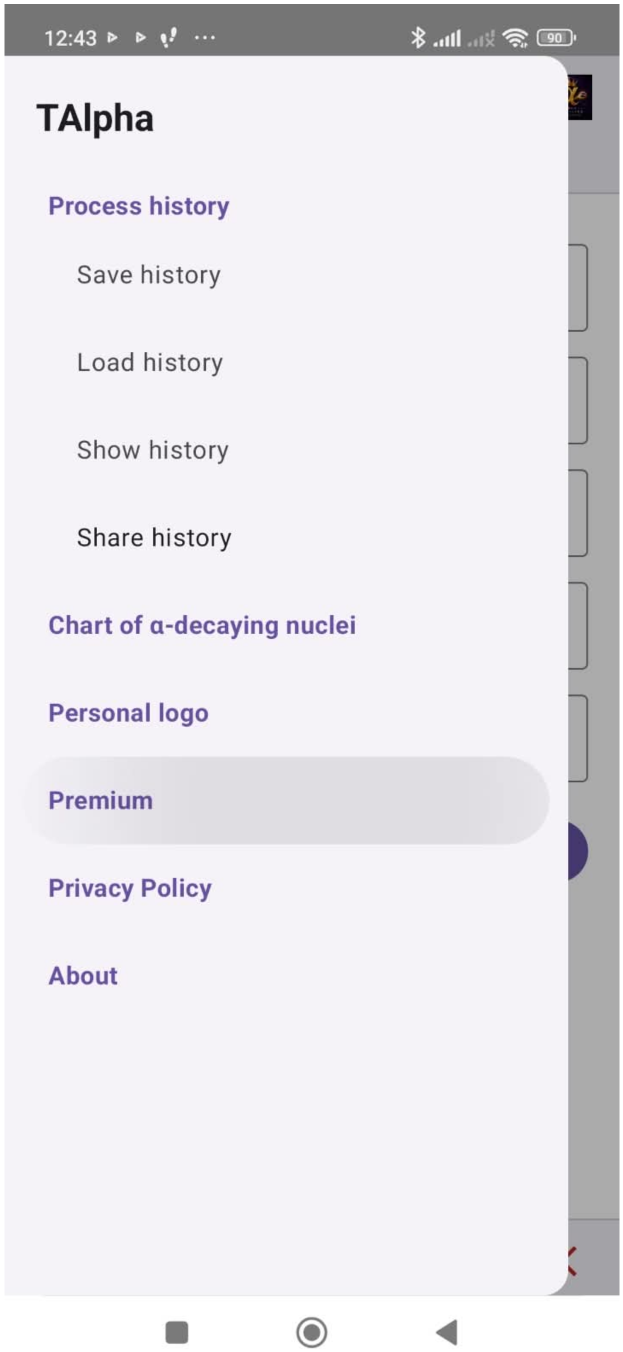

Figure 3. TAlpha side menu.

Process history

Save history. Saves the current calculation history to an internal file with a name chosen by the user.

Load history. Shows the list of previously saved files. The selected file is loaded as the current history; an individual file can be deleted with the red X.

Show history. Opens the current history and allows you to load one record into the calculator or delete it.

Share history. Allows you to select the current history or one saved file and send it through a supported application. This is a Premium feature.

Diagrams and settings

Chart of α-decaying nuclei. Opens the mass-number index and interactive diagrams for nuclides included in the current α-database.

Personal logo. A Premium setting for a personal square PNG logo no larger than 2 MB. A preview is shown after selection; the accepted logo is placed in the center of the bottom bar.

Premium. Opens the one-time Premium purchase screen. Premium removes advertisements and unlocks Chart, Share history and Personal logo. The purchase is associated with the app-store account and can be restored through the store.

Information

Privacy Policy. Opens the current privacy policy.

About. Shows information about TAlpha, its authors, scientific models, application version, and useful links.

> **If a menu item is unavailable.** Check Premium access for Share history and Personal logo. For web links and database downloads, check the connection. Cancelling a dialog does not change the current data.

History management

History management separates the current working set of records from saved files. This makes it possible to return quickly to an individual calculation, maintain several topic-specific sets, and share only the required file.

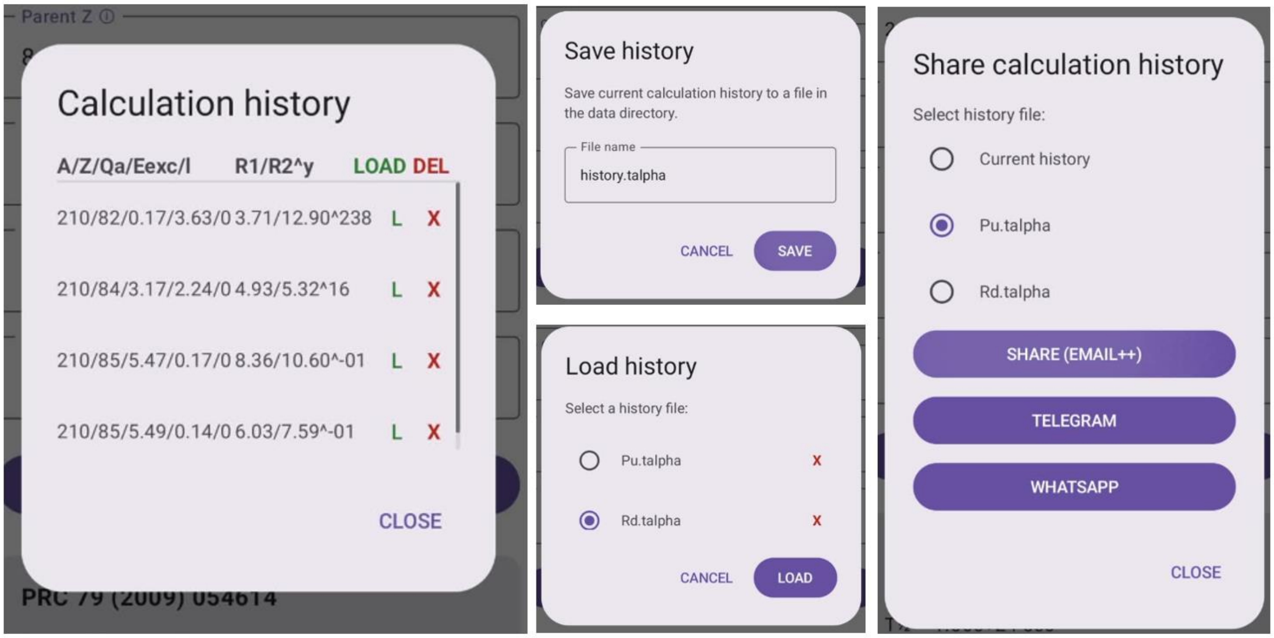

Figure 4. Left: current history; center: saving and loading files; right: selecting a history set to share.

Show history: current history

The Calculation history dialog contains calculation records and the following column groups and controls:

- A/Z/Qα/Eexc/l — the input parameters of the record;
- R1/R2^y — the results of the first and second calculation models in years;
- LOAD / green L — transfers the selected record to the main screen;
- DEL / red X — deletes one record after confirmation;
- CLOSE — closes the dialog without changing the remaining records.

If there are more rows than fit on the screen, scroll the list vertically. When deleting a record, the confirmation dialog shows its key parameters and any available experimental values, helping you avoid deleting the wrong calculation.

Loading one record

1. Open Menu → Show history.
2. Find the required row using A, Z, Qα, Eexc, and l.
3. Tap the green L in that row.
4. Check the populated fields on the main screen and tap CALCULATE. Loading a record does not run a new calculation automatically.

Save history: saving the current set

1. Open Menu → Save history.
2. Enter a meaningful name in the File name field. Keeping the .talpha extension makes the file easy for the application to recognize.
3. Tap SAVE. CANCEL closes the dialog without saving.

The complete current history is saved, including experimental fields when they are present in the records. The application manages the file format; it does not normally need to be edited manually.

Load history: loading and deleting files

1. Open Menu → Load history.
2. Select one file from the list using its radio button.
3. Tap LOAD to make the selected file the current history. CANCEL leaves the current history unchanged.

The red X to the right of a name deletes that saved file. Deleting a file and deleting an individual record are different operations: the first removes the entire saved set, while the second changes only the current history.

> **Before deleting a file.** If you may need the set later, first select it in Share history and save a copy in an external application, or load it and save it under another name.

Share history: sending a selected set (premium)

1. Open Menu → Share history.
2. Select Current history or a specific saved .talpha file.
3. Tap SHARE (EMAIL++), TELEGRAM, or WHATSAPP.
4. In the system dialog that opens, choose a recipient and confirm the send action. CLOSE exits the dialog without sharing.

The EMAIL++ button invokes the system share mechanism and may show email, cloud storage, and other compatible applications. The dedicated Telegram and WhatsApp buttons work when the respective application is installed and available on the device.

> **Privacy.** Before sharing, check the selected radio button: the application sends exactly the marked current history or saved file.

Chart of α-decaying nuclei (premium)

Chart of α-decaying nuclei is used to choose a nuclide from the database and navigate visually from the general index to an isobaric layer and then to the daughter-nucleus levels. Database version 3.0 and later combines two distinct kinds of information: potentially accessible levels and experimentally observed alpha transitions.

How to read the data. A level shown in the scheme is energetically accessible according to the adopted mass and level data. An experimentally identified branch is additionally marked by an arrow and may include T½exp. The absence of an arrow does not mean that the transition is forbidden.

General gestures and active elements

- Drag with one finger to move a magnified diagram across the screen;
- pinch or spread with two fingers to zoom out or in;
- tap a nuclide label or level to select that active element;
- a thin highlighted frame identifies the active area that registered the tap;
- SUMMARY and CHART move back by one logical level; CANCEL closes the index and returns to the calculator;
- use the system Back button/gesture for normal navigation when no dedicated on-screen button is shown.

After strong magnification, a label may move outside the screen. Zoom out or drag the canvas to bring it back. Zooming does not change the physical data or selected parameters.

Data shown in the diagrams

TAlpha uses a compact database in its own nuclide-data format. Each nuclide record stores A, Z, the element symbol, mass excess, Qα, and the estimated-mass flag; experimental T½ and normalized ENSDF data are added when available. The NDFS v3.0 extension adds adoptedLevels: evaluated daughter-nucleus levels that are energetically accessible for alpha decay.

- Qα and level energies are shown in keV on the diagrams and transferred to the calculator fields in MeV;
- T½exp is stored and displayed in years;
- lmin is calculated only when the parent and daughter Jπ assignments are sufficiently definite; if it cannot be determined reliably, no value is invented and the l label is omitted;
- experimental transitions are stored separately from the list of potential levels and do not replace it.

Summary

The first diagram screen is the Summary Scheme Index. It shows the available mass numbers A for which the loaded database contains layers.

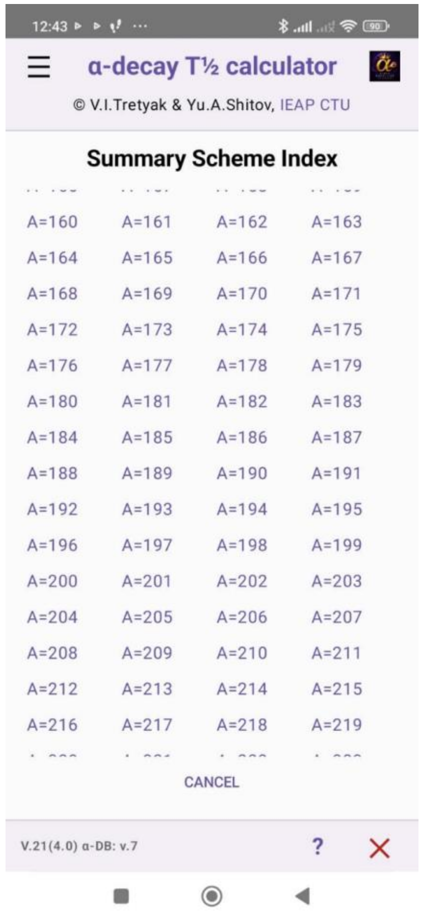

Figure 5. Mass-number index A.

Summary screen elements

A=<number>. An active link to the isobaric layer with the stated mass number. For example, A=210 opens the A=210 nuclides included in the current database.

Scrolling. Swipe up or down to reach A values that do not fit on the screen.

CANCEL. Closes the diagram section and returns to the calculator main screen.

The available A values and range boundaries are determined by the current α-DB version. The list on your device may therefore differ from the screenshot.

Opening a layer

1. Select a mass number A.
2. Wait for the A-level chart to be drawn. Rendering a large layer may take a short time.
3. If you selected the wrong A, tap SUMMARY on the next screen and choose another value.

A-level chart

The A-level chart shows nuclides in one isobaric layer A as a function of atomic number Z and relative energy. It is an overview screen for selecting a particular parent nuclide.

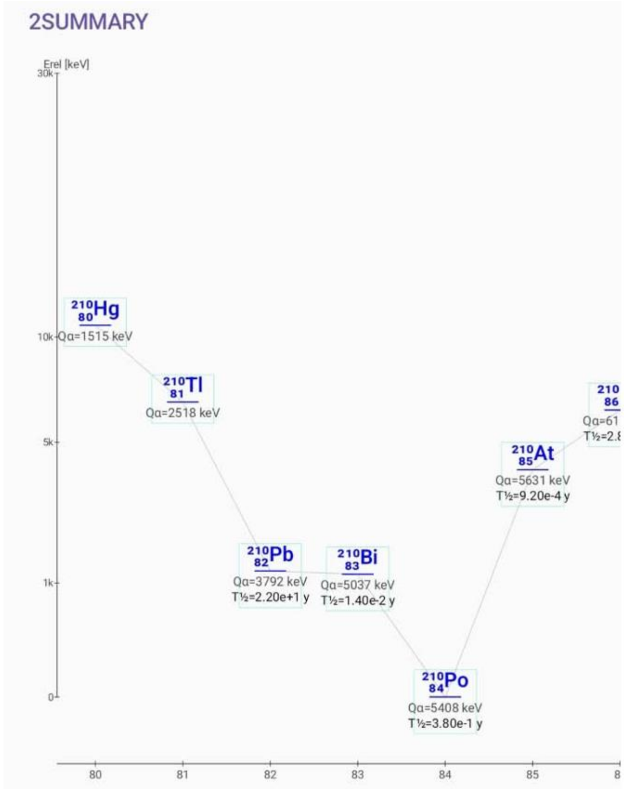

Figure 6. Example A-level chart for A = 210.

Axes and labels

- The horizontal axis is the atomic number Z;
- the vertical Erel [keV] axis is the relative energy position of the nuclide in the adopted layer representation;
- the blue nuclide notation contains the mass number A, element symbol, and atomic number Z;
- Qα is the ground-state-to-ground-state transition energy;
- T½ is the experimental half-life when available in the database;
- thin connecting lines help trace the sequence of points in the layer; they do not represent experimental alpha branches.

Active areas and navigation

Tap the label of the required nuclide. The active area is highlighted, and the Alpha-decay scheme for that nuclide opens. If the scheme does not open, make sure you tap inside the label frame and that data are available for the nuclide.

SUMMARY. Returns to the mass-number index without leaving the calculator's diagram section.

The chart can be zoomed and moved. This is especially useful when neighboring labels are close together or some isobars are near the edge of the screen.

Alpha-decay scheme

The Alpha-decay scheme shows decay of the selected parent nuclide to the ground and excited states of the daughter nuclide. It combines potential adoptedLevels with experimental ENSDF transitions while preserving their distinct meanings.

Figure 7. Expanded (LONG) scheme for the $^{210}At \rightarrow {}^{206}Bi$ alpha decay.

What the scheme shows

- The parent nuclide is at the upper right; its Jπ and ground-state marker (g.s.) may be shown next to it.
- The daughter nuclide is labeled at the bottom; the horizontal lines represent its ground and excited states.
- The excitation energy in keV and, when known, the spin/parity Jπ are shown to the left of a level.
- The Qα value for the specific branch is shown above or to the right of the line. For an excited level, it is Qα(gs→gs) − Eexc.
- The l label gives the minimum alpha-particle orbital angular momentum when it can be determined unambiguously from Jπ and the conservation laws. If Jπ is unknown or ambiguous, the l label is omitted.
- T½exp is shown for an experimental branch when the corresponding value is present in the database.

- A thin arrow from the parent nuclide is drawn only for an experimentally known alpha transition. A level line without an arrow is a potentially accessible final state.
- The vertical "Energy not to scale" label indicates that the fixed visual spacing between levels is not a linear energy scale.

SHORT and LONG

SHORT. A compact view for quickly selecting the most relevant low-centrifugal-barrier transitions. SHORT displays a limited set of levels that meet the application filter, including levels with small lmin; the current level-count limit is shown by the E_LEVEL_SHORT_MAX indicator.

LONG. An expanded view of all available levels that pass the database energy filter. Use zooming and panning when the scheme is long.

ⓘ. Opens an explanation of SHORT mode and the current E_LEVEL_SHORT_MAX limit.

Tapping SHORT/LONG changes only the presentation of the scheme. The nuclide data and experimental records remain unchanged.

Selecting a level and sending it to the calculator

1. If necessary, switch between SHORT and LONG and enlarge the required part of the scheme.
2. Tap the line or label of the selected daughter level.
3. The application transfers the parent A and Z, the selected branch Qα, Eexc, l, and T½exp to the calculator when these values are defined.
4. Check the parameters on the main screen. If necessary, supply an unknown l or adjust other values, then tap CALCULATE.

Selecting a level allows you to calculate a particular branch directly instead of only the ground-state transition. If lmin is not defined, the application must not assign a fictitious value; the user enters a physically justified l manually.

Returning to previous screens

CHART. Returns to the A-level chart for the selected mass number.

SUMMARY. When available on the current screen or after returning, opens the general A index.

After a level has been transferred to the calculator, you can return to the schemes through Menu → Chart of α-decaying nuclei. The application retains the selected context where supported by the current screen.

> **Interpretation limit.** TAlpha calculates a phenomenological estimate of T½. A potential level without an experimental arrow is a calculation candidate, not a claim that the branch has been observed. For scientific citation, consult the primary ENSDF evaluations and the publications describing the models.

Quick workflows

For a manual calculation:

1. Enter A, Z, Qα, Eexc, and l on the main screen.
2. Make sure that Qα − Eexc > 0, then tap CALCULATE.
3. Compare the two calculated-result cards and the EXP block, if shown.
4. Open Show history to return to the record, or Save history to save the set.

To select a transition from the database:

1. Open Menu → Chart of α-decaying nuclei.
2. Select A in Summary, then tap the required nuclide in the A-level chart.
3. In the Alpha-decay scheme, choose SHORT or LONG and tap the required level.
4. Check the transferred values on the main screen and run CALCULATE.

If a result looks unexpected

- Check that energies were entered in MeV rather than keV;
- compare the input Qα with the Qα on the EXP card—after a manual edit, EXP remains visible as a reference;
- check Eexc and confirm that Qα − Eexc is positive;
- make sure that l is the alpha-particle orbital angular momentum, not the daughter-level J;
- check the α-DB version in the bottom bar; if no version is shown, retry the database download using a stable connection;
- for a crowded diagram, zoom in and tap precisely inside the highlighted active area;
- remember that SHORT and LONG show different amounts of potential-level information.

Symbols and notation

Symbol	Meaning
A	Mass number of the parent nuclide
Z	Atomic number of the parent nuclide
Qα	Alpha-decay energy; for a branch, Qα = Qα(gs→gs) − Eexc
Eexc	Excitation energy of the daughter level
l / lmin	Alpha-particle orbital angular momentum / its minimum allowed value
Jπ	Spin and parity of a nuclear state
T½	Half-life
EXP	Experimental data
g.s.	Ground state
α-DB	Version of the loaded TAlpha database

Guide version. This document corresponds to the TAlpha v.4.2 interface and TAlpha NDFS v3.0 database format. The version and database numbers in the screenshots are examples and may differ from those installed on the device.